

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

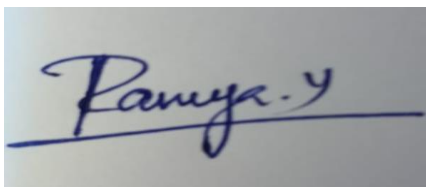
**Annexure A
APPLICATION BASED QUESTIONS**

Code of Conduct:

I Ramya. Y/ (192424397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a white background. The signature is written in a cursive style and appears to read 'Ramya. Y'. The signature is underlined with a single horizontal line.

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Ans: Probability that the blue ball came from Box C

$$P(A)=1/4, P(B)=1/2, P(C)=1/4.$$

$$P(\text{Blue}|A)=4/12=1/3, P(\text{Blue}|B)=7/15, P(\text{Blue}|C)=2/15.$$

$$P(C | \text{Blue}) = \frac{(2/15)(1/4)}{(1/3)(1/4) + (7/15)(1/2) + (2/15)(1/4)} = \frac{1/30}{7/20} = \frac{2}{21}$$

Answer: $P(\text{Box C} | \text{Blue}) = 2/21 \approx 0.0952$ (9.52%).

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Ans: Probability that the blue car came from Section C

$$P(A)=1/4, P(B)=1/2, P(C)=1/4.$$

$$P(\text{Blue}|A)=3/10, P(\text{Blue}|B)=6/15=2/5, P(\text{Blue}|C)=2/15.$$

$$P(C | \text{Blue}) = \frac{(2/15)(1/4)}{(3/10)(1/4) + (2/5)(1/2) + (2/15)(1/4)} = \frac{1/30}{37/120} = \frac{4}{37}$$

Answer: $P(\text{Section C} | \text{Blue}) = 4/37 \approx 0.1081$ (10.81%).

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary

classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

Ans: Breast cancer classification metrics

Total test samples = 20% of 3600 = 720, with 360 malignant and 360 benign.

Given TP=210, TN=190, FP=30.

FN = 360 – 210 = 150.

Metric	Value
Accuracy	$(210+190)/580 = 68.97\%$
Precision	$210/(210+30) = 87.50\%$
Recall (Sensitivity)	$210/(210+150) = 58.33\%$
Specificity	$190/(190+30) = 86.36\%$

Interpretation: High precision and specificity, but recall is moderate, so some malignant cases are missed.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

- (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
- (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
- (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
- (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

- i) Identify the epoch at which the model begins to overfit.
- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Ans: Loss values and overfitting

- Overfitting begins: Epoch 4 (validation loss starts increasing after reaching its minimum at epoch 3).
- Techniques: Early stopping and regularization (L2 or dropout).
- Effect: Overfitting reduces generalization ability; the model performs poorly on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

- i) Identify the epoch at which the model starts overfitting.
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance.

Ans: Accuracy values and overfitting

- Overfitting starts: Epoch 5 (validation accuracy stops improving and then decreases).
- Reason: The model memorizes training data, so training accuracy rises while validation accuracy falls.
- Improve validation performance: Use early stopping and regularization/dropout.

6. A machine learning model shows the following loss values during training:

- (a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
- (b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
- (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
- (d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
- (e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
- (f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5

(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8

(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1

- i) At which epoch does the model achieve optimal generalization?**
- ii) Identify signs of overfitting from the data.**
- iii) Recommend two regularization techniques and explain their impact.**

Ans: Optimal generalization from loss values

- Optimal generalization: Epoch 4 (minimum validation loss = 2.0).
- Signs of overfitting: Training loss continues decreasing while validation loss increases from epoch 5 onward.
- Regularization techniques: L2 regularization and dropout; both reduce model complexity and improve generalization.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).

- a) Front identified as Front: 119**
- b) Front identifies as Back: 06**
- c) Back identified as Back: 120**
- d) Back identified as Front: 05**

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Ans: Tire segregation metrics

TP=119, FN=6, TN=120, FP=5.

Metric	Value
Accuracy	$239/250 = 95.60\%$
Precision	$119/124 = 95.97\%$
Sensitivity	$119/125 = 95.20\%$
Specificity	$120/125 = 96.00\%$
F1-score	95.58%

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy):

- a) Diseased identified as Diseased: 138**

b) Diseased identified as Healthy: 12

c) Healthy identified as Healthy: 135

d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Ans: Medical diagnosis metrics

TP=138, FN=12, TN=135, FP=15.

Metric	Value
Accuracy	$273/300 = 91.00\%$
Precision	$138/153 = 90.20\%$
Recall	$138/150 = 92.00\%$
Specificity	$135/150 = 90.00\%$
F1-score	91.09%

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam.

Out of 400 emails (200 spam and 200 non-spam):

- **Spam identified as Spam: 180**
- **Spam identified as Not Spam: 20**
- **Not Spam identified as Not Spam: 170**
- **Not Spam identified as Spam: 30**

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Ans: Spam email filtering metrics

TP=180, FN=20, TN=170, FP=30.

Metric	Value
Accuracy	$350/400 = 87.50\%$
Precision	$180/210 = 85.71\%$
Recall	$180/200 = 90.00\%$
Specificity	$170/200 = 85.00\%$
F1-score	87.80%

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non- defective):

- **Defective identified as Defective: 230**
- **Defective identified as Non-Defective: 20**
- **Non-Defective identified as Non-Defective: 225**
- **Non-Defective identified as Defective: 25**

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Ans: Manufacturing quality inspection metrics

TP=230, FN=20, TN=225, FP=25.

Metric	Value
Accuracy	$455/500 = 91.00\%$
Precision	$230/255 = 90.20\%$
Recall	$230/250 = 92.00\%$
Specificity	$225/250 = 90.00\%$
F1-score	91.09%

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

Ans:Probability that the adulterated oil came from Shop Y

$P(X)=1/3$, $P(Y)=2/3$.

$P(\text{Adulterated}|X)=30/50=3/5$, $P(\text{Adulterated}|Y)=20/60=1/3$.

$$P(Y | \text{Adulterated}) = \frac{(1/3)(2/3)}{(3/5)(1/3) + (1/3)(2/3)} = \frac{2/9}{19/45} = \frac{10}{19}$$

Answer: $P(\text{Shop Y} | \text{Adulterated}) = 10/19 \approx 0.5263$ (52.63%).

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

Ans: Probability that the defective item came from Section A

$P(A)=3/4$, $P(B)=1/4$.

$P(\text{Defective}|A)=15/50=3/10$, $P(\text{Defective}|B)=25/50=1/2$.

$$P(A | \text{Defective}) = \frac{(3/10)(3/4)}{(3/10)(3/4) + (1/2)(1/4)} = \frac{9/40}{7/20} = \frac{9}{14}$$

Answer: $P(\text{Section A} | \text{Defective}) = 9/14 \approx 0.6429$ (64.29%).

13. In a binary classification problem using a deep learning model with a K- Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Ans: Binary classification metrics

Each class has 1000 samples; 20% test data gives 200 positives and 200 negatives.

$TP=120$, $TN=150$, $FN=200-120=80$, $FP=200-150=50$.

Metric	Value
Accuracy	$270/400 = 67.50\%$
Precision	$120/170 = 70.59\%$
Recall	$120/200 = 60.00\%$
Specificity	$150/200 = 75.00\%$

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity.

Ans: Medical image classification with KNN

Each class has 1200 samples; 25% test data gives 300 positives and 300 negatives.

$TP=180$, $TN=160$, $FN=300-180=120$, $FP=300-160=140$.

Metric	Value
Accuracy	$340/600 = 56.67\%$
Precision	$180/320 = 56.25\%$

Recall	$180/300 = 60.00\%$
Specificity	$160/300 = 53.33\%$

15. A deep learning model is trained for a classification task, and the following loss values are observed:

- i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6**
- ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2**
- iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9**
- iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8**
- v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9**
- vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1**
- vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4**
- viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7**

Answer the following:

- a. At which epoch does the model achieve the best generalization performance?**
- b. At which point does overfitting begin? Justify your answer.**
- c. Suggest two techniques to reduce overfitting and briefly explain how they help.**

Ans: Loss values and generalization

- Best generalization: Epoch 4 (minimum validation loss = 1.8).
- Overfitting begins: Epoch 5, because validation loss starts increasing while training loss continues decreasing.
- Techniques to reduce overfitting: Early stopping and regularization/dropout.